Graph-Enhanced Drug–Drug Interaction Detection

Application-Oriented Graph-Based Machine Learning on Cloud

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EECE 690 - Brief Comprehensive Report

Option A Graph: Yes

Metric	Warm Test	Cold-Start	vs. LR Baseline
AUROC	0.9738	0.9175	+0.0168
AUPR	0.9589	0.8824	+0.0115

Repository	`https://github.com/marwadeeb/Graph_DDI`
Live Demo	`https://marwadeeb-ddi-checker.hf.space/`
Dataset	DrugBank v5.1 (1)
Drugs	4,795 FDA-approved DDI pairs: 824,249

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1. Problem & Fit

1.1 Specific Problem

PF1

Adverse drug reactions (ADRs) are the fourth leading cause of death in the United States (2), responsible for over 125,000 deaths annually and 3–5% of hospital admissions (3). A primary contributor to ADRs is drug–drug interactions (DDIs): the modification of one drug's effect by the concurrent presence of another through shared metabolic enzymes, transporters, or pharmacological targets.

Problem statement: Given two FDA-approved drugs, does a clinically significant interaction exist between them? The task is formulated as *link prediction* on a heterogeneous pharmacological graph: predict whether an edge should exist between two drug nodes, given the topology of 824,249 documented DDI edges and 21,574 drug–protein target edges.

Inputs: two drug identifiers (name or DrugBank ID). **Output:** one of three states: `documented` (found in DrugBank Stage 1 lookup), `gnn_predicted` (not in DrugBank, GNN score ≥ 0.43), or `not_found`, together with a probability score and a pharmacological explanation.

1.2 User, Decision, and Deployer

PF2A

- **User:** Clinical pharmacists, physicians, and clinical decision-support system operators.
- **Decision being augmented:** Screening drug pairs for potential interactions prior to prescribing or dispensing.
- **Deployer:** Hospital pharmacy systems, EHR providers (Epic, Cerner), or formulary management platforms.
- **Payer / incentive:** Hospitals seeking to reduce adverse drug event liability, readmission rates, and associated costs.

1.3 Why ML Over a Simpler Approach

PF3A

A lookup table is sufficient for the 824,249 documented pairs, and we implement exactly that as Stage 1 (O(1) dictionary lookup). The ML component is justified for what a lookup cannot do:

1. **Coverage gap:** DrugBank lists 4,795 approved drugs, yielding $\binom{4795}{2} \approx 11.5\,\text{M}$ possible pairs. Only 7.2% are documented. A newly approved drug has *zero* lookup coverage.
2. **Cold-start performance:** Graph heuristics collapse to AUC ≈ 0.50 for cold drugs (zero training edges). Our GNN achieves 0.9175 by using drug feature embeddings and protein target structure.
3. **Mechanistic reasoning:** The GNN encodes 2,708 human protein targets (CYP450 enzymes, transporters, carriers) as a second node type; two drugs sharing CYP3A4 metabolism is a direct pharmacokinetic interaction signal unavailable to heuristics.

1.4 Impact and Significance

PF4

DrugBank (1) is acknowledged as incomplete, as many real interactions are not yet catalogued. A system capable of predicting novel interactions for the 93% of drug pairs absent from DrugBank has direct clinical value. The PubMedBERT encoder (9) grounds predictions in biomedical literature, and the protein target subgraph captures known pharmacokinetic mechanisms. The system is deployed publicly at `https://marwadeeb-ddi-checker.hf.space`.

1.5 Track Fit and Success Criteria

PF5

Type: Option A (application). **Graph:** Yes. Success criteria defined at project outset and achieved:

- AUROC ≥ 0.95 on held-out test set $\rightarrow$ Achieved: **0.9738**
- AUPR ≥ 0.93 $\rightarrow$ Achieved: **0.9589**
- Three-state pipeline verified end-to-end $\rightarrow$ Achieved
- Live deployed API $\rightarrow$ Achieved

2. Technical Rigor & Responsible ML

2.1 Task Formulation and Data

TM1

Task: Binary link prediction on an undirected drug–drug interaction graph under positive-unlabeled (PU) conditions: labeled positives are documented DDIs; unlabeled pairs may or may not interact.

Dataset: Starting from DrugBank v5.1 (1) (19,842 drugs, 2.9M directed DDI edges), we filter to 4,795 FDA-approved small molecules and 824,249 undirected DDI pairs. Protein target data is extracted from DrugBank's polypeptide tables, retaining 2,708 human proteins connected to drugs via 21,574 edges (targets, enzymes, transporters, carriers).

```
Dataset statistics:
  Drug nodes          : 4,795   (FDA-approved small molecules)
  DDI edges           : 824,249 (undirected, from DrugBank)
  Protein nodes       : 2,708   (human proteins from DrugBank)
  Drug->protein edges: 21,574  (targets, enzymes, transporters, carriers)
  Drug feature dim    : 980     (212 structural + 768 PubMedBERT)
  Protein feature dim: 5       (4 role flags + log-degree)
  Prior pi            : 0.0717  (= 824249 / C(4795,2))
```

2.2 Non-AI Baselines

TM2A

Five non-AI baselines are evaluated on the identical test split (Table 1). *Feature Cosine Similarity* and *Degree Product* use no graph topology. *Common Neighbors*, *Adamic–Adar* (10), and *Jaccard Coefficient* exploit DDI graph structure.

Table 1: Non-AI and non-graph baselines on the warm test set.

Method	AUROC	AUPR
Feature Cosine Similarity (no graph)	0.6041	0.5952
Degree Product	0.9534	0.9383
Common Neighbors (11)	0.9738	0.9648
Adamic–Adar (10)	0.9748	0.9660
Jaccard Coefficient	0.9763	0.9669
Logistic Regression (no graph)	0.9570	0.9474
GNN — HeteroGraphSAGE + NCN (ours)	**0.9738**	**0.9589**

Key findings: Feature Cosine (pure non-AI, no graph) scores 0.6041; our GNN exceeds it by +0.3697 AUROC. Notably, topology-based heuristics (e.g., Jaccard, Adamic–Adar) slightly outperform the GNN on the warm evaluation. This is expected given the high density of the DDI graph, where most drug pairs share common neighbors. However, these methods fundamentally fail in cold-start settings, where drugs have no observed interactions. Therefore, while heuristics are competitive in retrospective evaluation, they are unsuitable for the intended deployment scenario involving newly approved or sparsely connected drugs.

2.3 Method Choice and Substantive Contribution

TM3

Three architectural decisions distinguish our model from a vanilla GNN baseline.

2.3.1 Heterogeneous Graph Augmentation

The original homogeneous drug–drug graph is augmented with 2,708 human protein nodes and 21,574 drug–protein edges, creating a heterogeneous graph processed by `HeteroConv` (8) in PyTorch Geometric (7). This adds pharmacokinetic signal unavailable in the DDI graph alone (Table 2).

2.3.2 NCN-style Common Neighbor Pooling

Inspired by Neural Common Neighbor (NCN) (6), our `EnhancedLinkPredictor` decoder replaces the standard endpoint concatenation with a pooled representation of shared DDI neighbors and shared protein targets:

$$\mathbf{h}_{uv} = \mathrm{MLP}\big([\mathbf{z}_u \,\|\, \mathbf{z}_v \,\|\, \mathrm{mean}_{w\in\mathcal{N}(u)\cap\mathcal{N}(v)}(\mathbf{z}_w) \,\|\, \mathrm{mean}_{p\in T(u)\cap T(v)}(\mathbf{z}_p)]\big)$$

where $\mathcal{N}(\cdot)$ denotes DDI graph neighbors and $T(\cdot)$ denotes protein targets. Input dimension: $64 + 64 + 64 + 64 = 256$.

2.3.3 nnPU Learning

Because DrugBank is incomplete, most "negative" pairs are unlabeled rather than confirmed non-interacting. Standard binary cross-entropy would treat unlabeled pairs as hard negatives, introducing systematic label noise. We use the non-negative Positive-Unlabeled risk estimator (5):

$$\tilde{R}_{\mathrm{nnPU}}(g) = \pi \hat{R}_P^+(g) + \max\big(0,\ \hat{R}_U^-(g) - \pi \hat{R}_P^-(g)\big)$$

with prior $\pi = 0.0717$ (observed positive rate). The $\max(0, \cdot)$ clamp prevents risk from going negative with flexible models (5).

Table 2: Architecture ablation on the held-out test set.

Model	AUROC	AUPR
Homo GraphSAGE (V1)	0.9615	0.9450
Hetero GraphSAGE V1 (no CN pooling)	0.9726	0.9586
Hetero + EnhancedDecoder (ours)	**0.9738**	**0.9589**

2.4 Preprocessing, Features, and Leakage Control

TM4

Drug features (980-dim):

- *Structural [0:212]:* Morgan fingerprints (radius 2), ring counts, hydrogen bond donors/acceptors, TPSA, MW, logP, rotatable bonds, and CYP450 substrate flags (3A4, 2D6, 2C9, 2C19, 1A2). Computed via RDKit (14).
- *Semantic [212:980]:* 768-dim PubMedBERT (9) embeddings of drug description text, capturing pharmacological class and mechanism of action.

Protein features (5-dim): $[\mathbb{1}_{\text{target}}, \mathbb{1}_{\text{enzyme}}, \mathbb{1}_{\text{transporter}}, \mathbb{1}_{\text{carrier}}, \text{log-degree}]$. Role flags are multi-hot; log-degree is normalized by maximum degree.

Leakage controls:

- 20% of DDI edges masked before any split (`maskRatio=0.2`): masked edges never seen during training.
- `RandomLinkSplit` (7) applied to remaining 80%, producing non-overlapping train/val/test sets.
- Node features computed from drug chemistry before any split; no label information leaks into features.
- Negative sampling during training draws exclusively from pairs not in the positive set.

2.5 Evaluation Protocol

TM5

```
Split configuration:
  Total DDI edges (directed):  1,648,498
  Masked (nnPU, never trained):  329,699  (20%)
  Training edges:              1,055,564
  Validation edges:              131,944  (10% of visible)
  Test edges:                    131,944  (10% of visible)
```

Metrics: AUROC and AUPR (average precision). Both are threshold-free and appropriate for imbalanced binary classification. AUPR is the primary metric for PU learning settings, where precision at high confidence matters most for clinical utility.

Mini-batch training: Stochastic mini-batch sampling via `NeighborLoader` (7) with neighbor counts [25, 15, 10] for DDI edges and [10, 5, 3] for drug–protein edges. Batch size: 512 drug seed nodes.

2.6 Error Analysis

TM6

Table 3: Confusion matrix at threshold $\tau = 0.43$ on the test set.

	Pred. Positive	**Pred. Negative**
Actual Positive	TP: 55,406	FN: 10,566
Actual Negative	FP: 3,055	TN: 62,917

Precision: 0.9477. When the model flags an interaction, it is correct 94.8% of the time. This is the clinically critical direction: false alarms lead to unnecessary dose adjustments and patient anxiety.

Degree analysis:

Table 4: Average combined DDI degree by error category.

Category	Avg. Combined Degree
True Positives	2,291.9
False Positives	1,851.0
False Negatives	2,167.6

The degree gap between FN (2168) and TP (2292) is small, indicating the model struggles with specific interaction patterns rather than purely data-sparse drugs. FP pairs involve moderately-connected drugs that share structural/topological similarity without documented DDIs. This is an expected behavior in PU learning where some "negatives" are undocumented positives.

2.7 Limitations and Trade-offs

TM7

- DrugBank labels are incomplete; true negatives are uncertain (PU learning assumption holds but introduces estimation error in π).
- 2,107/4,795 drugs have no protein target data; their embeddings rely solely on DDI graph topology.
- Trained on FDA-approved small molecules only; generalization to biologics or prodrugs is untested.
- GNN predictions are graph-structural, not mechanistic: a high score reflects similarity to known interaction patterns.
- Threshold $\tau = 0.43$ selected on validation; clinical deployment requires prospective calibration.
- Transductive setting: the model cannot generalize to drugs completely absent from the training graph.

Prior experiments confirm that raising the prior to $\pi = 0.12$ (V4) does not improve performance (AUROC 0.9692 vs. 0.9738), and adding JK-Net residuals (V3) yields marginal gains (AUROC 0.9733) at higher complexity; both alternatives were discarded.

2.8 Graph as Core Modeling Object

TM9G

The graph is not auxiliary, but is the primary modeling object. **Nodes:** 4,795 drugs (980-dim features) and 2,708 proteins (5-dim features). **Edges:** (drug, `ddi`, drug), (drug, `targets`, protein), (protein, `rev_targets`, drug). The encoder is a 3-layer `HeteroConv` (8) propagating messages along all three edge types. The decoder explicitly pools embeddings of shared graph neighbors. There is no model without the graph.

2.9 Graph vs. Non-Graph Baseline and Cold-Start Evaluation

TM10G

A logistic regression (12) trained on raw 980-dim drug features (no graph, pair features = $[|\mathbf{f}_A - \mathbf{f}_B|, \mathbf{f}_A \odot \mathbf{f}_B]$) provides the non-graph ML baseline. The cold-start evaluation holds out 10% of drugs entirely (zero training edges), simulating newly approved drugs entering the formulary.

Table 5: Warm and cold-start comparison.

Setting	Method	AUROC	AUPR
Warm	Logistic Regression	0.9570	0.9474
	GNN (ours)	**0.9738**	**0.9589**
	Δ	+0.0168	+0.0115
Cold-start	Heuristics (all)	0.5000	0.5000
	Logistic Regression	0.8974	0.9032
	GNN (ours)	**0.9175**	**0.8824**

Cold-start is the GNN's structurally guaranteed advantage. Graph heuristics score analytically zero for cold drugs (no training neighbors $\Rightarrow$ all heuristic scores = 0, AUC = 0.50). Logistic Regression recovers to 0.8974 using node features alone. The GNN exceeds LR by +0.0201 AUROC by combining node features with partial graph context, confirming that graph structure adds real signal beyond raw drug chemistry.

2.10 Explainability

RM1

Feature group ablation: We zero out one feature group at inference time and measure AUROC drop (Table 6). Structural features (CYP450 flags, fingerprints) contribute 10× more than PubMedBERT embeddings, confirming that pharmacological structure, not drug name semantics, drives prediction.

Table 6: Feature group ablation on the test set:

Variant	AUROC	AUPR	Δ AUROC
Full features (980-dim)	0.9738	0.9589	-
Ablate structural [0:212]	0.8993	0.8549	−0.0745
Ablate PubMedBERT [212:980]	0.9661	0.9464	−0.0077

CN pooling ablation: Zeroing each decoder component isolates the contribution of structural pooling (Table 7). The marginal warm-eval drop reflects that all test drugs are well-connected; the pooling provides most signal in sparse/cold-start settings.

Table 7: Common neighbor pooling ablation on the test set.

Decoder variant	AUROC	AUPR	Δ AUROC
Full decoder	0.9738	0.9589	-
Ablate shared DDI neighbours	0.9719	0.9563	−0.0019
Ablate shared protein targets	0.9717	0.9562	−0.0021
Ablate both	0.9724	0.9576	−0.0014

Per-prediction explanations: The `explain()` function in `gnn_predictor.py` surfaces shared

enzyme, transporter, and target counts with plain-English reasons displayed on the live web interface for every GNN prediction.

2.11 Fairness and Bias

RM2

Table 8: AUROC stratified by subgroup.

Subgroup	AUROC	n pairs
High-degree drugs ($\geq$ median)	0.8844	66,066
Low-degree drugs ($<$ median)	0.9867	65,878
Both drugs have protein targets	0.9159	74,341
At least one lacks targets	0.9930	57,603

Counter-intuitively, low-degree and protein-uncovered drugs score *higher*. High-degree hub drugs interact with many partners, creating genuine ranking ambiguity at test time. Drugs with protein target data are pharmacologically complex (multiple CYP450 roles) and their interactions are harder to predict. The model is challenged by *pharmacological complexity*, not data sparsity.

2.12 Privacy and Leakage

RM3

Data source: DrugBank Full Database v5.1 (CC BY-NC 4.0, academic license). No patient-level data is used at any stage. Leakage controls: 20% edge masking before splitting; strict non-overlapping train/val/test sets via `RandomLinkSplit`; node features computed before any split; negative sampling restricted to non-positive pairs. The deployed model accepts only DrugBank IDs and returns probability scores. No patient data is processed. Input sanitization strips control characters and enforces maximum request length.

2.13 Robustness and Distribution Shift

RM4

Table 9: Robustness under two forms of distribution shift.

Perturbation	AUROC	AUPR	Δ AUROC
No perturbation	0.9738	0.9589	-
Edge dropout 20%	0.9714	0.9555	-0.0024
Edge dropout 40%	0.9717	0.9559	-0.0021
Feature noise $\sigma = 0.1$	0.9703	0.9534	-0.0035
Feature noise $\sigma = 0.5$	0.9283	0.8779	-0.0455

Removing 20–40% of DDI edges causes < 0.003 AUROC drop, which is critical for clinical deployment, where real-world databases are incomplete. Feature noise at $\sigma = 0.5$ (extreme, well beyond typical descriptor variability) causes a 0.045 drop, the expected failure mode for a feature-dependent model.

3. Deployment & Engineering

3.1 Dockerized API

EN1

The application is fully containerized. A `docker-compose.yml` provides one-command local deployment:

```
cp .env.example .env   # add GROQ_API_KEY
docker compose up --build -d
# App available at http://localhost:7860
curl http://localhost:7860/health
```

The `Dockerfile` installs all Python dependencies, copies data files (tracked via Git LFS), and starts the Flask (13) app via Gunicorn on port 7860. The same image is deployed to HuggingFace Spaces.

3.2 Separation of Data, Model, and Serving

EN2

```
parser/              <- DrugBank XML parsing -> 27 normalized CSVs
pipeline/            <- feature engineering, graph construction, training
  hetero_model.ipynb      <- GNN training (standalone, reproducible)
  gnn_predictor.py        <- inference layer (loaded once, stateless)
  step5_hetero_graph.py   <- heterogeneous graph construction
  step9_baseline.py       <- baseline evaluation
data/                <- all artifacts (CSVs, .pt graphs, evaluation JSONs)
app.py               <- Flask serving layer (calls gnn_predictor.predict())
templates/           <- HTML frontend
```

`gnn_predictor.py` exposes a clean public API: `predict(id_a, id_b) -> dict` and `explain(id_a, id_b) -> dict`. `app.py` imports it as a black box. Training, inference, and serving are fully decoupled.

3.3 Reproducible Environment

EN3

`requirements.txt` pins all dependencies. Full pipeline:

```
python parser/run_all.py                  # ~2 min: XML -> 27 CSVs
python pipeline/dedup_interactions.py     # ~3.5 min
python pipeline/filter_approved.py        # ~20 s
python pipeline/build_graph.py            # ~15 s: 212 structural features
python pipeline/embed_drugs.py            # ~25 min (CPU): PubMedBERT
    embeddings
python pipeline/build_pyg_hetero.py       # ~30 s: hetero graph with proteins
# GNN training: pipeline/hetero_model.ipynb (~17 min on GPU)
python pipeline/step9_baseline.py         # ~30 s: all baselines + cold-start
python pipeline/responsible_ml.py         # ~5 s: bias + robustness
```

`hetero_model.ipynb` contains all data loading, split, model definition, training loop, and evaluation in a single runnable notebook with embedded outputs.

3.4 Functional UI / Demo Flow

EN4

Live at `https://marwadeeb-ddi-checker.hf.space`. Six pages serving distinct functions:

Table 10: Web application pages.

URL	Description
`/`	Landing page with animated pipeline and drug graph visualization
`/checker`	Drug pair checker: autocomplete, three-state results, GNN explanation
`/chat`	NLP chatbot: Groq LLM NER extracts drug names from free text
`/results`	Model performance: cold-start + full baseline comparison
`/responsible`	Responsible ML: explainability, fairness, privacy, robustness
`/dashboard`	Live dashboard: query stats, system health, recent activity

Demo pairs for each output state:

- `documented`: Warfarin + Aspirin (DB00682 + DB00945) — in DrugBank, Stage 1 instant match.
- `gnn_predicted`: Vincristine + Sonidegib (DB00541 + DB09143) — not in DrugBank, GNN score 0.8271.
- `not_found`: Bicalutamide + Asenapine (DB01128 + DB06216) — not in DrugBank, GNN score 0.4006.

3.5 Running Deployed Artifact

EN5

Deployed and publicly accessible at `https://marwadeeb-ddi-checker.hf.space` via HuggingFace Spaces (Docker SDK). The space auto-rebuilds on every push to the main branch. Example API call:

```
curl -X POST https://marwadeeb-ddi-checker.hf.space/api/check \
  -H "Content-Type: application/json" \
  -d '{"drug_a": "Vincristine", "drug_b": "Sonidegib"}'
# Response:
# { "source": "gnn_predicted", "found": true,
#   "gnn": {"probability": 0.8271},
#   "explanation": {"reasons": ["Both drugs share 2 metabolic enzyme(s)
    ..."]},
#   "disclaimer": "...consult a pharmacist..." }
```

4. GitHub & Documentation

4.1 Repository Structure

GD1

Repository: `https://github.com/marwadeeb/Graph_DDI`

```
Graph_DDI/
|-- data/
|   |-- step3_approved/   <- 27 normalized DrugBank CSVs
|   |-- step4_graph/      <- ddi_graph.pt, hetero_ddi_graph.pt,
    bestHeteroModel.pt
|   `-- evaluation/       <- baselines_results.json, responsible_ml_*.json
|-- docs/                 <- ARCHITECTURE, model_arch, api_reference, ...
|-- parser/               <- DrugBank XML parsing pipeline (12 modules)
|-- pipeline/             <- ML pipeline (12 scripts + hetero_model.ipynb)
|-- templates/            <- Flask HTML (6 pages)
|-- app.py, Dockerfile, docker-compose.yml
|-- requirements.txt, README.md
`-- readme_correction.md  <- rubric code -> exact file/page location
```

4.2 README and Setup

GD2

`README.md` covers: quick start (Docker Compose), full pipeline commands, API usage examples, key results table, page index, documentation index, data source attribution, and deployment notes. `readme_correction.md` maps every rubric code to the exact file and page.

4.3 Method and Architecture Documentation

GD3

The `docs/` directory contains dedicated files:

- `docs/model_architecture.md`: GNN architecture, 980-dim feature breakdown, ablation results.
- `docs/pipeline.md`: step-by-step pipeline, run commands, file structure.
- `docs/ARCHITECTURE.md`: tech stack index across all components.
- `docs/api_reference.md`: all REST endpoints with request/response format.
- `docs/data_schema.md`: 27 normalized DrugBank tables, key statistics.
- `docs/responsible_ml.md`: RM1–RM4 analysis with numbers.

4.4 Results, Logs, and Ablations

GD4

`pipeline/hetero_model.ipynb` contains all experiment results with embedded cell outputs: results tables (warm + cold-start + prior-work comparison), architecture ablation (Table 2), feature group ablation (Table 6), CN component ablation (Table 7), error analysis (Table 3), fairness analysis (Table 8), and robustness analysis (Table 9).

`data/evaluation/` contains machine-readable JSON outputs from `step9_baseline.py` and `responsible_ml.py` rendered as tables on the `/results` and `/responsible` pages.

4.5 Data Sources, Limitations, and Deployment Notes

GD5

- **Data:** DrugBank Full Database v5.1, CC BY-NC 4.0. Raw XML not redistributed; derived CSVs tracked via Git LFS.
- **Limitations:** Documented in `hetero_model.ipynb` (§TM7) and `docs/responsible_ml.md`.
- **Disclaimer** on all GNN predictions: "This is a clinical decision support signal, not a confirmed finding. Consult a pharmacist or clinician before use."
- **Security:** CSP headers, X-Content-Type-Options, Referrer-Policy, input sanitization (strip control chars, max length), request size cap (512 KB).

5. Creativity & Initiative

CI1 CI2 CI3 CI4

Originality: The combination of heterogeneous drug–protein graph + NCN-style CN pooling + nnPU learning on the same model is an original architecture. No prior work applies NCN-style structural pooling to heterogeneous DDI graphs with protein bipartite edges.

Design trade-offs:

- *nnPU prior:* sensitivity analysis ($\pi \in \{0.0717, 0.12\}$) confirms the observed positive rate is the best estimate. JK-Net + residuals (V3: AUROC 0.9733) and higher prior (V4: AUROC 0.9692) were both evaluated and discarded.
- *Warm vs. cold-start:* warm eval favors heuristics due to graph density; cold-start is the honest test for the deployed use case. Both are reported transparently.

Beyond minimum requirements:

- AI chatbot (`/chat`) with Groq LLaMA NER for free-text drug name extraction and plain-English DDI explanation.
- Per-prediction pharmacological explanation via `explain()`: surfaces shared CYP450 enzymes, transporters, and target counts with plain-English reasons.
- Live dashboard (`/dashboard`) with real-time query statistics and system health.
- Cold-start evaluation framework, not a standard rubric item, added to honestly characterize model behavior.
- Security hardening: CSP headers, input sanitization, request size limits.

6. Overall Results Summary

Table 11: Full results table including prior-work comparison.

Setting	Model	AUROC	AUPR
Warm	Feature Cosine Similarity	0.6041	0.5952
	Degree Product	0.9534	0.9383
	Common Neighbors	0.9738	0.9648
	Adamic–Adar	0.9748	0.9660
	Jaccard	0.9763	0.9669
	Logistic Regression	0.9570	0.9474
	Homo GraphSAGE V1	0.9615	0.9450
	Hetero GraphSAGE V1	0.9726	0.9586
	Hetero + EnhancedDecoder (ours)	**0.9738**	**0.9589**
Cold-start	Heuristics	0.5000	0.5000
	Logistic Regression	0.8974	0.9032
	GNN (ours)	**0.9175**	**0.8824**
Prior work	LLM-DDI (15)	0.9571	0.7346
	ACDGNN (16)	0.9835	0.9881
	Ours	**0.9738**	**0.9589**

Notes: ACDGNN operates on 500–1,500 drugs with confirmed negatives, which is a substantially easier setting than our 4,795 drugs with PU learning. We exceed LLM-DDI on AUPR by +0.2243. Feature Cosine (pure non-AI, no graph): 0.6041; our GNN exceeds it by +0.3697 AUROC.

7. Rubric Summary

Table 12: All applicable rubric items for Option A + Graph: Yes.

Code	Criterion	Evidence
PF1	Specific problem	§1.1 + DrugBank lookup + GNN pipeline
PF2A	User/decision/deployer	§1.2
PF3A	Why ML	§1.3 (3 arguments)
PF4	Impact	§1.4 + live deployment
PF5	Track fit + success criteria	§1.5 (all 4 criteria met)
TM1	Task + data formulation	§2.1
TM2A	Non-AI baselines	§2.2 (5 baselines, Table 1)
TM3	Method choice	§2.3 (3 contributions)
TM4	Preprocessing/leakage	§2.4
TM5	Evaluation protocol	§2.5
TM6	Error analysis	§2.6 (Tables 3 and 4)
TM7	Limitations	§2.7
TM9G	Graph core object	§2.8
TM10G	Graph vs. non-graph	§2.9 (Table 5)
RM1	Explainability	§2.10 (Tables 6 and 7)
RM2	Fairness/bias	§2.11 (Table 8)
RM3	Privacy/leakage	§2.12
RM4	Robustness	§2.13 (Table 9)
EN1	Dockerized API	§3.1 + Dockerfile
EN2	Separation	§3.2
EN3	Reproducible env	§3.3 + requirements.txt
EN4	UI/demo	§3.4 + `marwadeeb-ddi-checker.hf.space`
EN5	Running artifact	§3.5 + curl example
GD1	Repo structure	§4.1
GD2	README	§4.2 + readme_correction.md
GD3	Method docs	§4.3 + docs/
GD4	Results/ablations	§4.4 + hetero_model.ipynb
GD5	Data/limits/notes	§4.5
CI1–CI4	Creativity	§5

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